

Standardized AI Solutions Framework (SAS-F)

**Built for engineering &
delivering enterprise grade
AI use-cases**

Patent pending

Introduction

The Standardized AI Solutions Framework (SAS-F) is designed to provide a comprehensive and structured approach to engineering and delivering artificial intelligence (AI) services. SAS-F empowers enterprises to accelerate AI adoption at scale, combining customized use cases with the processing power of high-performance GPU servers to drive business breakthroughs.

SAS-F aims to address this need by providing a standardized framework that encompasses the entire AI service delivery lifecycle, from data gathering and model training to deployment and management. By leveraging SAS-F, clients can tap into the full potential of AI and accelerate their digital transformation journeys.

FIRST STAGE

Bespoke Solution Engineering



1. Services Definition Phase

Define the scope and objectives of the AI project, identifying key stakeholders and their requirements.



3. Solution Inception and Discovery

Conduct a thorough analysis of the business problem, identifying opportunities for AI-driven solutions and defining the solution vision.



5. Data Modalities

Evaluate and assess the modalities of data and fit them into structured, unstructured and semi-structured buckets.



7. Solution Architecture Design

Design a comprehensive architecture for the AI solution, integrating data ingestion, processing, and deployment.



9. Microservices Architecture Planning

Identify the AI frameworks and microservices involved for the identified AI workflow architecture.



2. Business Requirements Elicitation

Gather and document business needs and requirements through stakeholder interviews, surveys, and workshops.



4. Data Readiness Assessment

Evaluate the quality, completeness, and relevance of existing data sources to determine their suitability for architecting the AI solution.



6. Data Architecture Design

Design a robust data architecture that supports the integration of multiple data sources and ensures data governance.



8. AI Workflow Architecture

Evaluate and identify if the solution architecture requires an inference-based workflow - agentic workflow, RAG based workflow or involves fine-tuning/training.



10. Platform Selection and Configuration

Select and configure the most suitable infrastructure and hardware platforms to support AI solution and deployment.

SECOND STAGE

Data Curation & Centralization



1. Data Ingestion and Integration

Implement a scalable data ingestion framework from the end-customer systems to integrate disparate data sources, ensuring seamless data flow and synchronization.



3. Data Quality Assessment

Assess the quality of the data by implementing data validation, data cleansing, and data normalization techniques.



5. Data Warehouse and Lake Formation

Design and implement a centralized data warehouse and lake architecture that enables storing, processing, and analyzing large volumes of structured and unstructured data.



2. Data Pipeline Development

Develop and deploy data pipelines that streamline data processing of ingested data enabling efficient data movement and reduced latency.



4. Database Implementation

Create a robust and optimized database schema ensuring data consistency, scalability, and performance.

THIRD STAGE

AI Solution Development



1. Model selection and training

Curated data and its attributes define the ultimate set of precise and accurate models applicable for the engineering workflows. Identified models are bucketized to directly be inferred, be few-shot prompt engineered, or trained and fine-tuned, on a case-by-case basis, while meeting the key quality metrics for scalability, interpretability, and transparency in model performance and decision-making.



3. DAPT

Curated data amassing to a significant token count builds a solid foundation for continued Domain Adaptive Pre-Training over existing foundation models. Pipelines developed output models with state-of-the-art domain-specific text generation capabilities, analytical reasoning and problem-solving capabilities.



5. API Design and Development

Develop and implement APIs that enable secure, scalable, and standardized data exchange between AI and ML systems, applications, and services, promoting interoperability and reusability.



2. Pipeline Development

Sophisticated pipelines developed taking the customer's final interests into account, primarily to handle data flowing through one or many sub-functions: Message Queuing and Stream Handling functions, Dimensionality Reduction and Stratification functions, Embedding and Vector Database functions, SQL/NoSQL Database functions, Agentic/ Tool Calling functions, Evaluation Functions and more.



4. Application Development and Implementation

Design and develop applications that leverage AI and ML solutions, ensuring seamless integration with existing systems and infrastructure.

FOURTH STAGE

Guard-railing, Evaluation and Validation



1. Output Control and Guard-railing

Implement toxicity and bias detection, factuality checks, and hallucination filtering to ensure safe and accurate responses. Functions to enforce response length limits, restrict sensitive topics, and provide citations for verifiable information engineered. Post-processing pipelines on rule-based filtering, confidence scoring, and leveraging external moderation APIs developed to refine outputs before delivery.



2. Quality Assurance (QA) and Testing

Develop and execute comprehensive testing strategies to ensure AI and ML solutions meet stringent quality and performance standards, to identify and address defects and vulnerabilities.



4. Performance Benchmarking and Optimization

Measure, analyze, and optimize the performance of AI and ML models and the overall solution, identifying bottlenecks and areas for improvement to ensure scalability, accuracy, and cost-effectiveness.



3. Model Validation and Verification

Evaluate and validate AI and ML models involved in the solution against business requirements and performance metrics, ensuring accuracy, reliability, and fairness in model outputs and decision-making.



5. Security and Compliance Auditing

Conduct thorough security audits and risk assessments to ensure the solution complies with regulatory requirements, industry standards, and organizational RBAC policies, protecting sensitive data and preventing potential threats and vulnerabilities.

FIFTH STAGE

Deploy, Manage and Continuous Improvement



1. Platform Provisioning

Provision and manage infrastructure and resources to support AI and ML solutions and associated applications, ensuring scalability, security, and cost-effectiveness.

a. Accelerated Compute Infrastructure

Plan, design and implement scalable infrastructure solutions like VMs with GPUs or vGPUs, Kubernetes and Slurm Clusters, physical servers etc.



b. Networking

Deploy scalable networking as per OEM's reference architecture, ensuring low latency.



c. Storage

Storage interfaces like S3, Block and Shared FS are implemented to meet the high-throughput and low-latency.



d. Scaling

Automatic scaling of workloads to ensure the AI/ML solutions are operating as the demand increases.



e. Clustering solutions

Deploying solutions like Kubernetes, Slurm, which are adhered to OEM's architecture to enable training and inferencing of AI solutions.



f. Monitoring

Deploy and maintain monitoring solutions capabilities across IaaS, PaaS and SaaS layers. This includes deployment & management of tools which monitor the compute and application (API) performance.





2. Solution Deployment and Release Management

Coordinate and execute the deployment of AI and ML solutions and associated applications, ensuring seamless release management, minimal downtime, and optimal system performance.



3. Monitoring and Observability

Provide monitoring and observability capabilities across IaaS, PaaS and SaaS layers.



4. Transition and Handover

Facilitate a seamless transition of AI and ML solutions and associated applications from development to production, ensuring thorough knowledge transfer, documentation, and handover to operational teams.



5. Post-Implementation Review

Conduct a comprehensive review of AI and ML solution implementations, assessing project outcomes, identifying lessons learned, and documenting best practices to inform future initiatives.



6. Continuous Monitoring and Improvement

Proactively monitor AI and ML solutions in production, identifying areas for improvement, and implementing enhancements to ensure ongoing performance optimization, quality, and business value.

a. MLOps

Engineering and deploying end-to-end MLOps pipelines for seamless model deployment, monitoring, and scaling in production.



b. LLMOps

Optimizing and managing large language model deployments with fine-tuning, version control, and scalable infrastructure.



c. Inferencing Latency Optimization

Implementing low-latency AI solutions using quantization, batching, caching, and hardware acceleration for real-time performance. Parallely, monitoring the performance of production systems, with important performance metrics like average response time, error rates, system health and more.



7. Service Level Agreement (SLA) Management

Develop, negotiate, and manage SLAs that define service quality, availability, and responsiveness expectations for AI and ML solutions, ensuring alignment with business objectives and customer needs.

Conclusion

In conclusion, the Standardized AI Solutions Framework (SAS-F) represents a significant milestone in our efforts to deliver cutting-edge AI services that meet the evolving needs of our clients. By standardizing our approach to AI service delivery, we can ensure consistency, quality, and reliability across all our client engagements.

The SAS-F framework is designed to be flexible, adaptable, and scalable, allowing us to respond quickly to changing market conditions and client requirements. As we continue to innovate and expand our AI offerings, the SAS-F framework will remain a vital component of our strategy, enabling us to drive business growth, improve client satisfaction, and maintain our position at the forefront of the AI services industry.

By adopting the SAS-F framework, we are committed to delivering exceptional value to our clients, empowering them to unlock the full potential of AI and achieve their business goals. We believe that the SAS-F framework will become a key differentiator for our organization, setting us apart from competitors and establishing us as a trusted partner for businesses seeking to harness the power of AI.

About NxtGen:

NxtGen is India's largest sovereign cloud and AI provider, serving 1000+ customers across the country. NxtGen operates SpeedCloud® - public cloud platform, SpeedCloud Supreme – an alternative to private cloud, and Industry vertical clouds for the Government and Financial Services. NxtGen delivers value to its customers by ensuring superior performance at lower cost. NxtGen AI portfolio unifies GPU platforms from NVIDIA, AMD & Intel and provides a wide choice of highly optimized models to build enterprise use cases. NxtGen is working to truly democratize the cloud and AI technologies, by making them viable for enterprises of all sizes.

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